

8

The RCS of Two-Dimensional Parabolic Antennas

8.1 THE MAJOR SCATTERING COMPONENTS

The following investigation is limited to the parabolic cylinder with a feed placed at the focal line. However, the scattering mechanism of a parabolic dish with a typical feed is easily understood by simple extension of the cylindrical case.

The backscatter of a parabolic antenna consists essentially of two components:

- a. The scattering coming directly from the reflector itself without any involvement from the feed. We shall call it the *reflector scattering*.
- b. The scattering from the feed as the incident signal is focused upon it and reradiated via the reflector. We will denote this the *feed scattering*.

8.1.1 The Reflector Scattering

In Fig. 8.1 we show the bistatic scattering pattern for a parabolic cylinder with aperture width equal to 14λ and without any object at the focal line. In other words, this constitutes the reflector scattering alone. The cylindrical reflector is modeled by placing 96 columns of reactively loaded dipoles along a parabolic surface. Thus, the surface acts like an FSS surface similar to our approach shown in Figs. 5.2 and 5.3. This will ensure minimal leakage through the reflecting surface as opposed to using just infinitely long rods (see Sections 5.2 and 5.3). The SPLAT program was eminently well-suited for calculating these configurations.

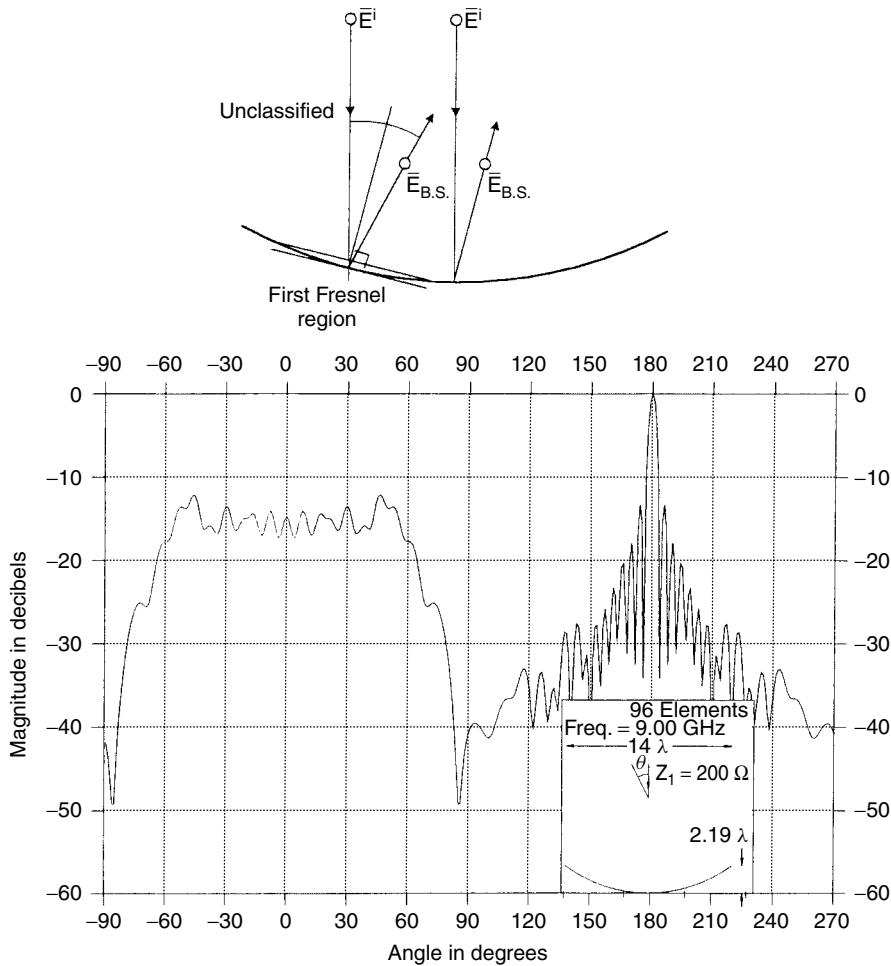


Fig. 8.1 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, at $\theta = 0^\circ$. There is no feed; that is, this shows the scattering from the parabolic surface itself referred to as reflector scattering.

The general shape of the bistatic pattern in the back sector is readily understood by noting that the scattered field is essentially coming from the first Fresnel zone associated with the particular bistatic directions as shown at the top of Fig. 8.1.

The field in the forward sector can basically be obtained approximately by integration of the incident field over the aperture width as discussed in Section 2.9.2. Note, however, that the entire pattern in Fig. 8.1 was obtained from the SPLAT program and therefore is as exact as the method of moments. In other words, the effect of edge currents is rigorously observed. See also Section 8.5.

8.1.2 Total Scattering from a Parabolic Reflector with a Typical Feed

The total scattering from a parabolic reflector with a typical feed is more complicated. As we shall see, it depends very much on the type of feed being used as well as upon the circumstances. Basically we can assume that one of the strongest backscatter cases is obtained when the feed deteriorates into a single column of resonating lossless dipoles placed at the feed line. This case is shown in Fig. 8.2 for incidence along the boresight axis. We clearly recognize the main beam at $\theta = 0^\circ$ while the reflector scattering in Fig. 8.1 has been broken up by interference from the feed scattering.

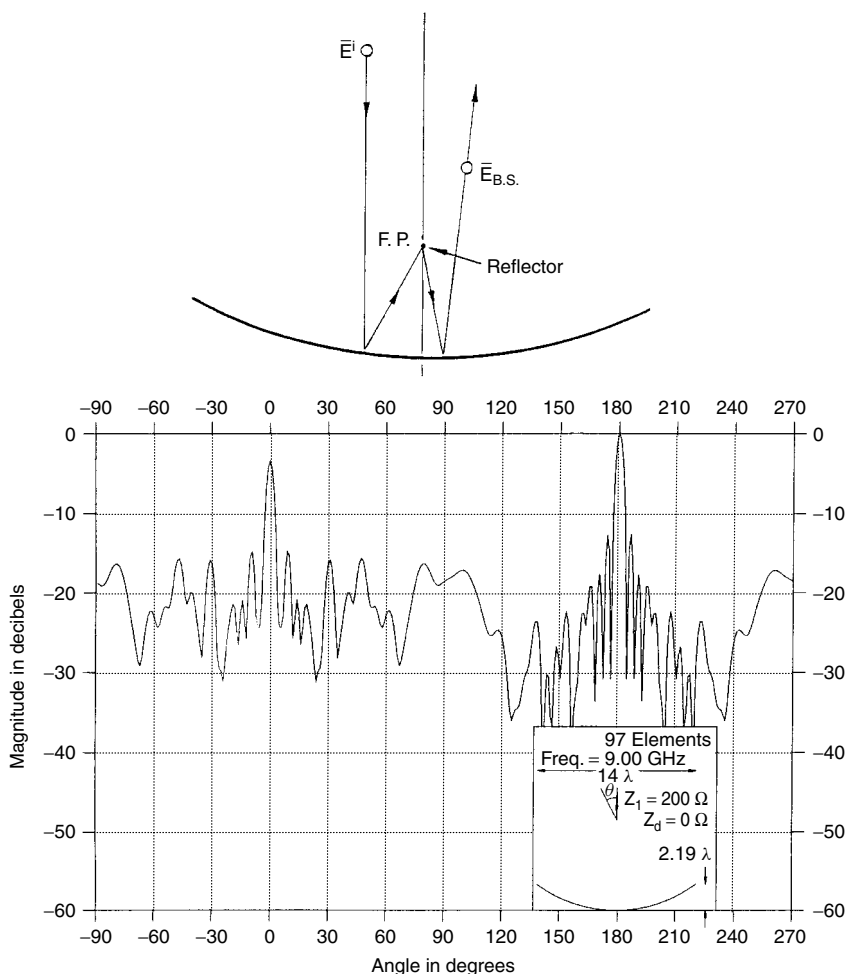


Fig. 8.2 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight — that is, at $\theta = 0^\circ$. There is a collinear array of short-circuited dipoles at the focal line, resulting in a large backscatter in the boresight direction.

If an actual feed in the form of a flared parallel plate waveguide is used, the backscatter can be significant but in general not as strong as the resonating dipole case in Fig. 8.2—that is, as long as we are incident along the boresight axis. If we are obliquely incident upon the reflector such that the incident field is focused on one of the edges of the horn we will, for the E field parallel to the edges, obtain a very strong return similar to the resonating dipole case in Fig. 8.2. Such a case is shown in Fig. 8.3. We clearly see a main beam in the bistatic pattern a few degrees to the right. At this angle of incidence the monostatic return coincides with the bistatic return as indicated in Fig. 8.3, top. This situation is of more than theoretical interest. It is typically encountered in practice during acquisition, leading to large RCS values on each side of the boresight axis. It therefore is of utmost importance to design a new type of feed without edges. This can be accomplished by simple modification of the omnidirectional antenna discussed earlier in Chapter 7.

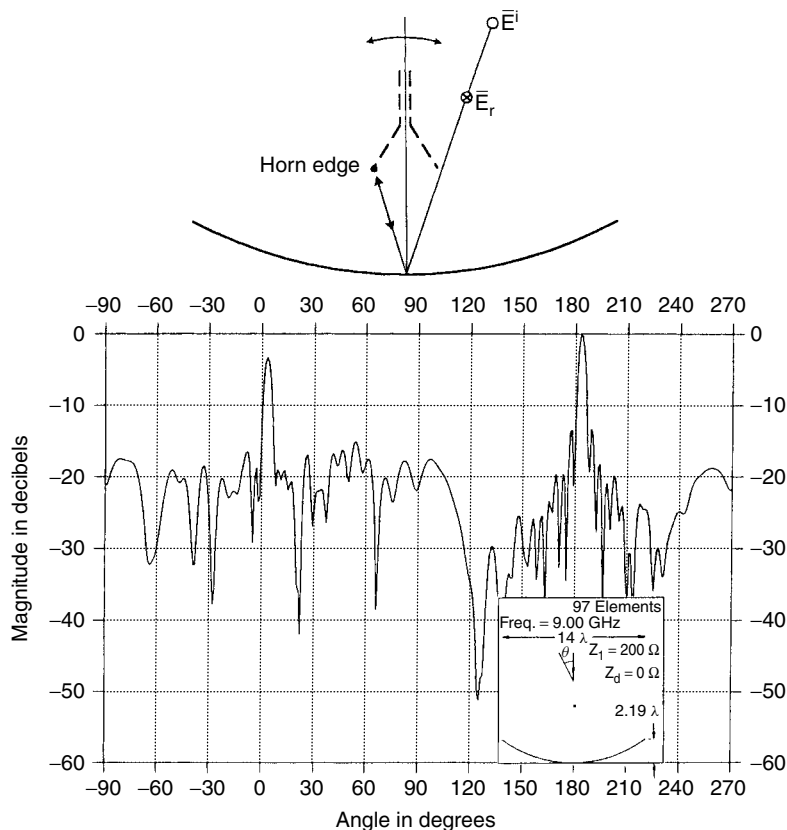


Fig. 8.3 The bistatic scattered field from a parabolic cylinder with a wedge-shaped "horn." When the incident field is slightly off boresight, the incident energy will focus at the edge of the feed, resulting in a large backscatter as shown.

8.2 TOTAL SCATTERING FROM A PARABOLIC REFLECTOR WITH LOW RCS FEED

Horn antennas have long been popular in the microwave community, and rightly so. They can simply be considered natural extensions of waveguides. However, the walls in horns are not necessarily needed to direct the energy in the forward direction as some intuitively might expect. This job can be handled in a number of other ways—for example, by using arrays. The beauty of this approach is simply that unwanted scattering from the walls can be avoided. To this end we shall start with the omnidirectional antenna discussed in Chapter 7 by simply exciting only part of the elements such that a suitable illumination of the reflector is obtained. The rest of the elements (or at least part of them) would be resistively loaded and serve to absorb the incident energy either directly or via the reflector.

The active elements would of course be fed from hybrids as explained in Section 2.12 and in detail in Section 8.3. Thus, from a scattering point of view we merely have to consider an omnidirectional antenna placed at the focal line with all the elements resistively loaded.

It should be admitted that some energy could be lost in the passive element because they may be excited by the active elements. However, we intend to excite the active elements in a tapered fashion, which should lead to minimum loss in the passive elements. For practical execution of such a feed, see Section 8.3.

A typical bistatic scattering pattern for a cylindrical parabolic reflector with a low RCS feed is shown in Fig. 8.4. The incidence is along the boresight axis at the center frequency 9.0 GHz. Also shown for comparison is the bistatic scattering pattern for the same reflector but without any feed—that is, the same as shown earlier in Fig. 8.1. Much to many readers' surprise, we observe that the scattering with the low RCS feed is about 6 dB lower than that with no feed at all. An explanation for this apparent paradox is illustrated in Fig. 8.4, top. In the no-feed case to the left, the incident signal will be focused on the focal line from where it continues into the back sector and will thus be incorporated into the reflector scattering, whereas the low RCS feed will simply absorb the energy converging into the focal area and thus not be incorporated into the reflector scattering.

As stated earlier, the results shown in Fig. 8.4 are obtained for incidence along the boresight axis and at the center frequency 9.0 GHz. Obviously our suggested low RCS feed should be tested at other frequencies as well as at other angles of incidence. This will be the subject of discussion in the next two sections.

8.2.1 The Bistatic Scattered Field as a Function of Angle of Incidence

Figures 8.5 through 8.9 show the bistatic scattered field for a parabolic cylinder with a low RCS feed. The frequency is fixed at the center frequency 9.0 GHz

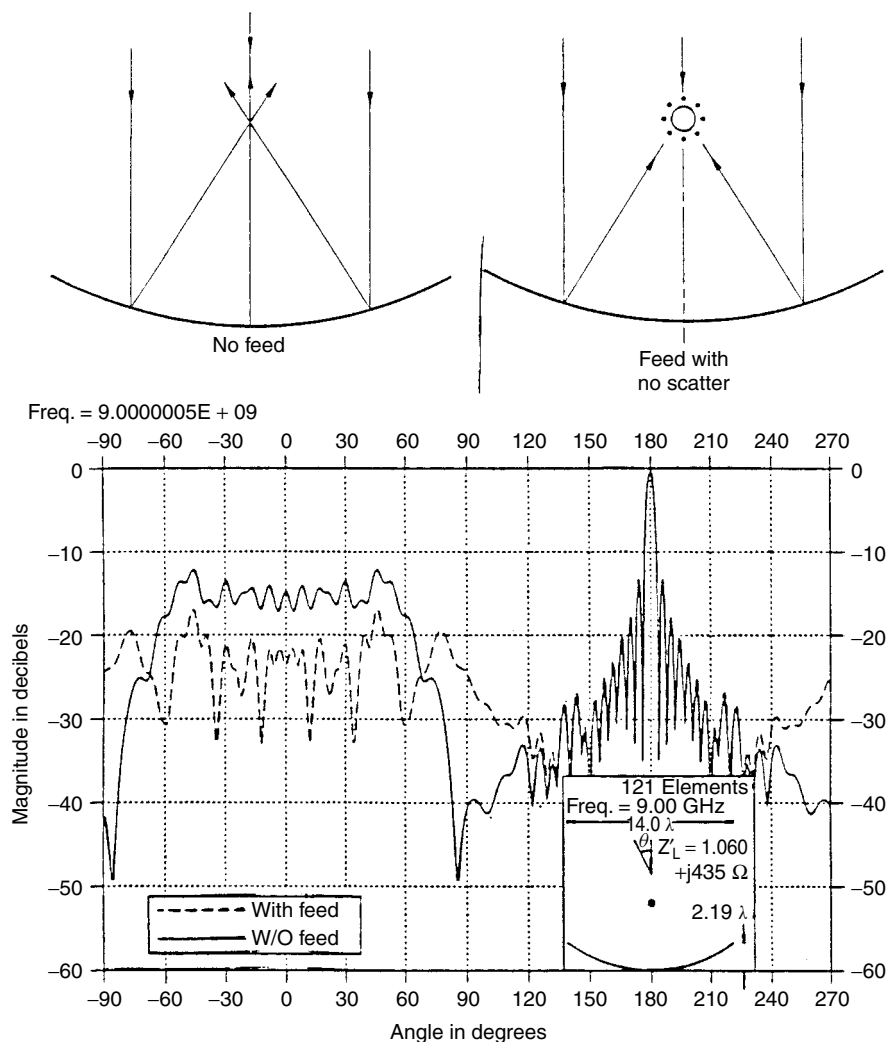


Fig. 8.4 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, $\theta = 0^\circ$. Shown with and without a low RCS feed. Frequency is 9.0 GHz.

while the angles of incidence are $\theta = 1.25^\circ, 2.5^\circ, 3.75^\circ, 5.0^\circ$, and 10.0° , respectively. These angles were chosen based on certain criteria. At $\theta = 1.25^\circ$ the central ray is just clearing the central pipe of the low RCS feed as indicated in the drawing at the top of Fig. 8.5. Note that the angles are not drawn to scale, nor is the size of the low RCS feed. This enables us to better assess the relationship among the incident field, the scattering, and the feed itself. This “distortion” of the figures will be observed in all the subsequent figures to follow.

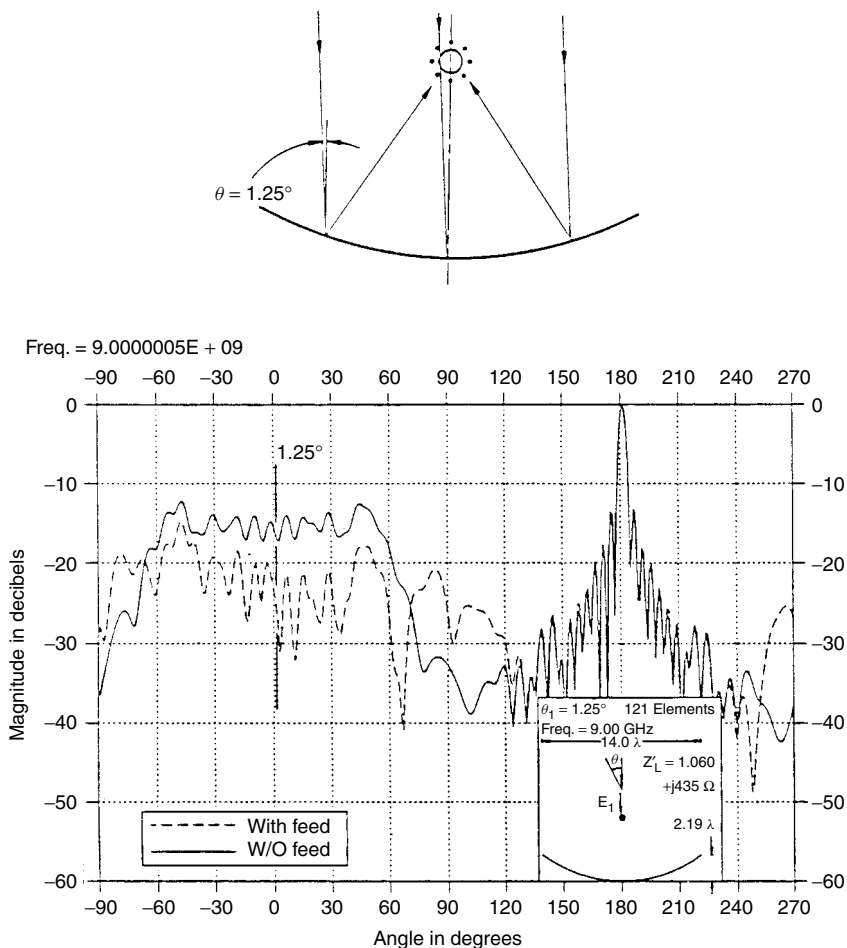


Fig. 8.5 The bistatic scattered field from a parabolic cylinder for an incident plane wave arriving at $\theta = 1.25^\circ$. Shown with and without low RCS feed. Frequency is 9.0 GHz.

Similarly, we show in Fig. 8.6 the bistatic scattered field when the incident signal is at 2.5° from boresight. We observe in this case that the incident direction just clears the location circle of the dipoles while we in Fig. 8.7 with $\theta = 3.75^\circ$ are well outside this circle. At $\theta = 5.0^\circ$ and 10.0° shown in Figs. 8.8 and 8.9, respectively, we clear the feed to the extent that we basically obtain reflector scattering only. All in all this “movie” shows a gradual change from $\theta = 0^\circ$ to $\theta = 10^\circ$ without any particular surprises.

8.2.2 The Bistatic Scattered Field as a Function of Frequency

We next show a series of calculated bistatic scattered patterns where the angle of incidence is fixed at 0° along the boresight axis while we vary the frequency

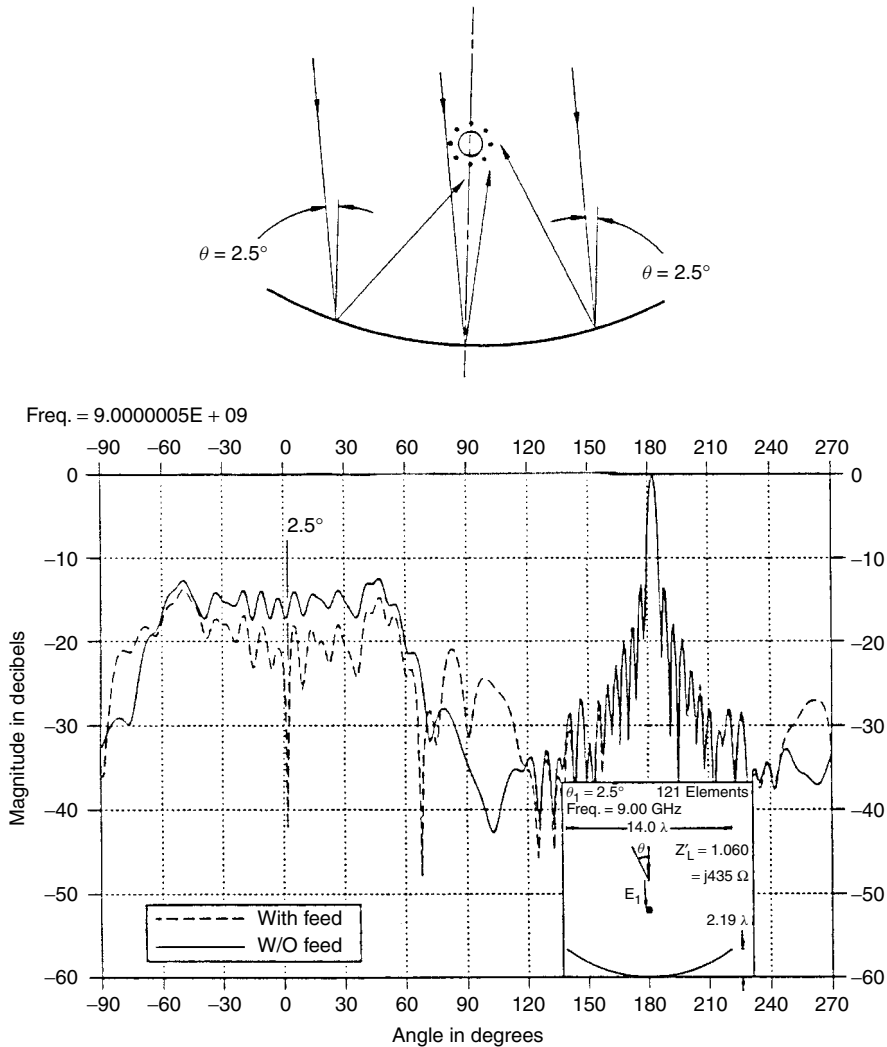


Fig. 8.6 The bistatic scattered field from a parabolic cylinder for an incident plane wave arriving at $\theta = 2.5^\circ$. Shown with and without low RCS feed. Frequency is 9.0 GHz.

downward as well as upward from the center frequency 9.0 GHz. More specifically, we show in Figs. 8.10 through 8.13 the bistatic scattered pattern at the frequencies 8.5, 8.0, 7.5, and 7.0 GHz, respectively. Similarly, we show in Figs. 8.14 and 8.15 the pattern at 9.5 and 10.0 GHz, respectively. We note that for all of these patterns the scattering with the low RCS feed is consistently below that of the parabolic reflector without feed. In other words, the low RCS feed works very efficiently over a significant band.

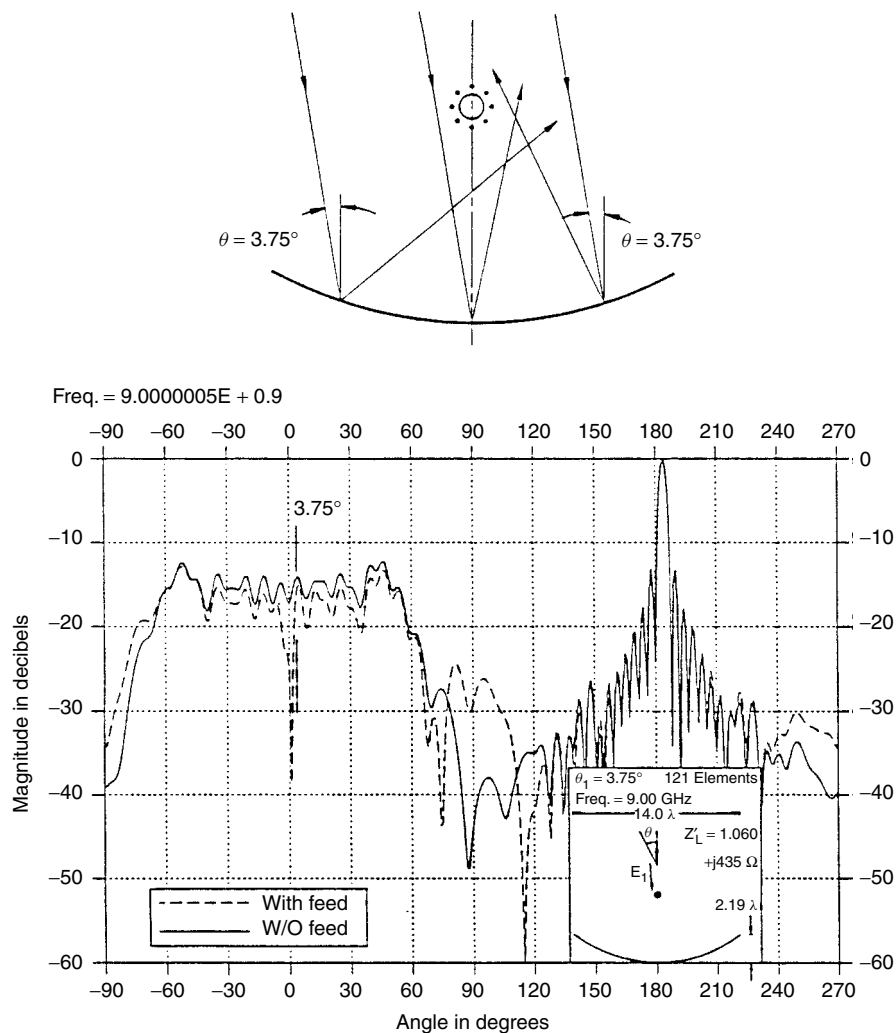


Fig. 8.7 The bistatic scattered field from a parabolic cylinder for an incident plane wave arriving at $\theta = 3.75^\circ$. Shown with and without low RCS feed. Frequency is 9.0 GHz.

8.3 PRACTICAL EXECUTION OF THE LOW RCS FEED

A possible execution of a low RCS feed arrangement is shown in Fig. 8.16. To the left we show the transmitting case and to the right the receiving case. We show four dipoles being feed via hybrids with power division $\sim 2\text{--}4\text{ dB}$. This will under transmitting condition produce a tapered aperture distribution as desired. (In reality, more dipoles and stronger taper may be needed.) However, when receiving, all dipoles are basically being exposed to the same power level. The tapered illumination will again be created by the hybrids with the excess

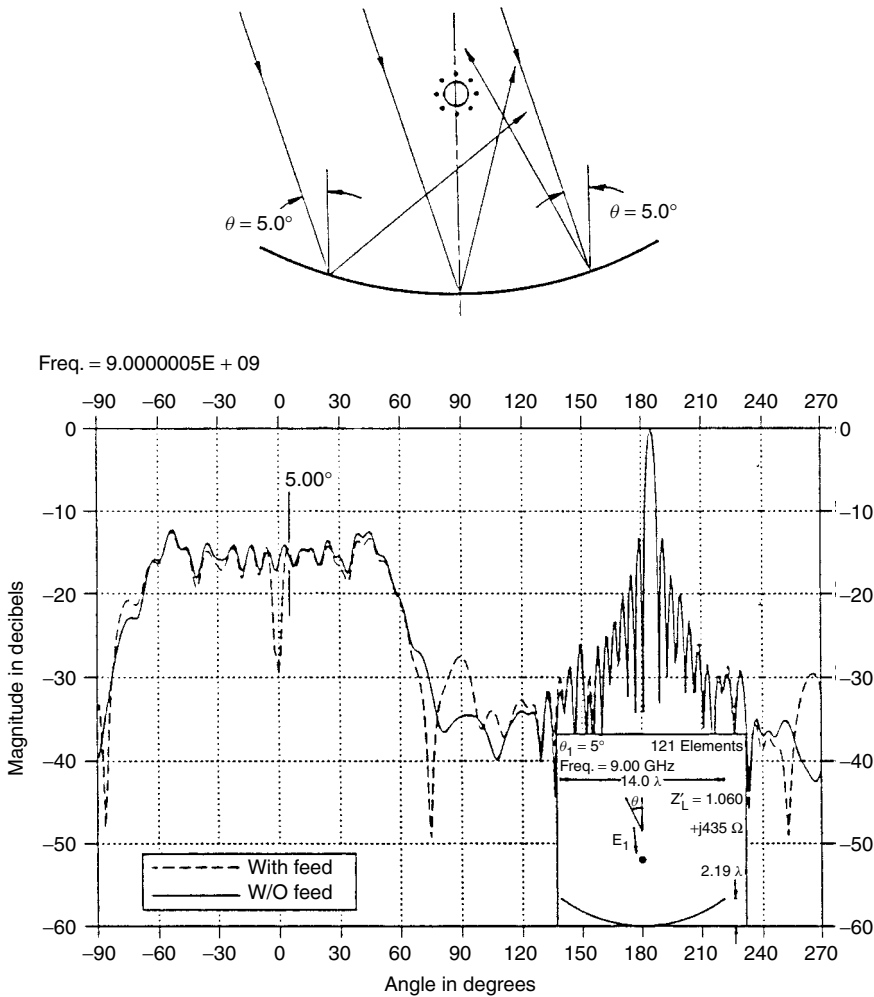


Fig. 8.8 The bistatic scattered field from a parabolic cylinder for an incident plane wave arriving at $\theta = 5.0^\circ$. Shown with and without low RCS feed. Frequency is 9.0 GHz.

received power being absorbed by the hybrid loads. That is, no power will be reradiated. Further discussion about this mechanism is given in Section 2.12 under planar arrays.

8.4 OUT-OF-BAND REDUCTION

We have so far considered RCS in-band reduction only. The success of this approach depends strongly upon how well the antenna is matched to the load (i.e., the terminal impedances of receivers or transmitters). Since a good match

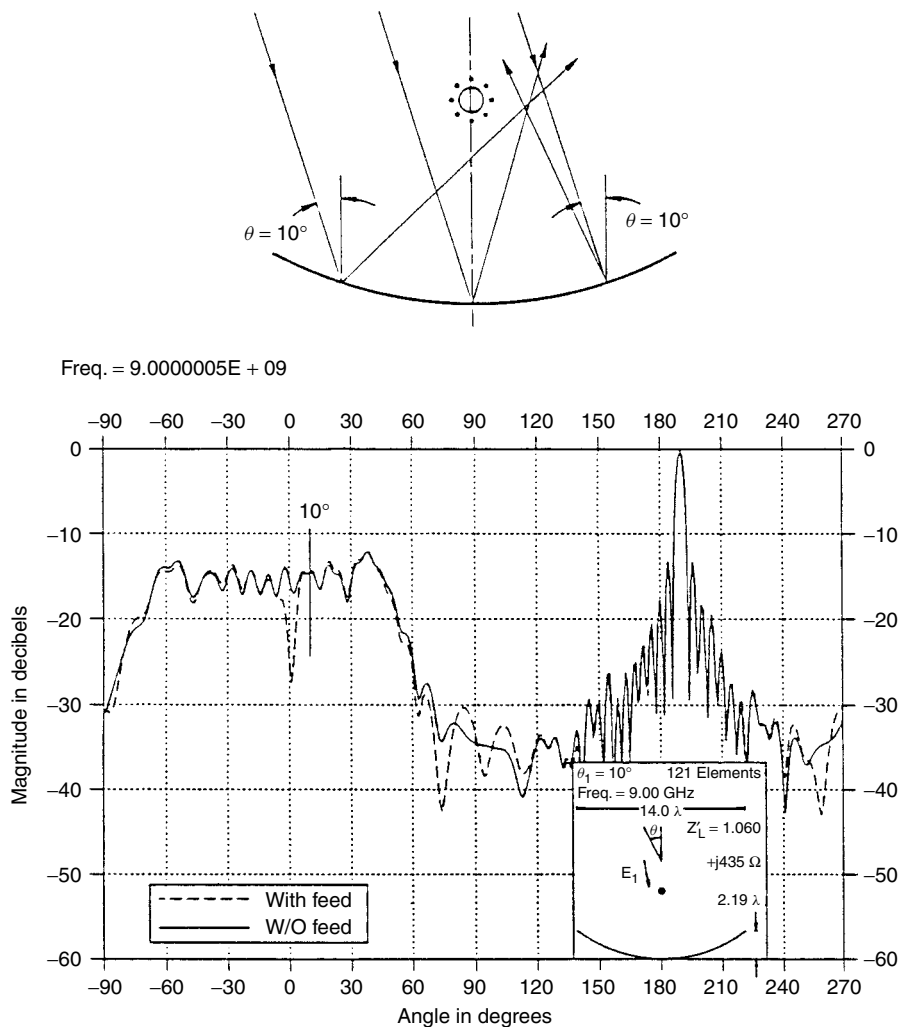


Fig. 8.9 The bistatic scattered field from a parabolic cylinder for an incident plane wave arriving at $\theta = 10.0^\circ$. Shown with and without low RCS feed. Frequency is 9.0 GHz.

is difficult to maintain over a large frequency band, we often resort to other approaches outside the operating band like the use of bandpass radomes (see Fig. 2.1).

One design covers the entire reflector. An example is given in Fig. 8.17. It is extremely effective but it requires a large, heavy, and expensive radome. The good news is that it in general also reduces the wind resistance.

Another design covers only the feed itself. An example is shown in Fig. 8.18. It is important that the wedge angle (or cone angle) is correct such that energy is not reflected back toward the reflector nor in the backscatter direction. Also, the

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 Freq. = 8.5000003E + 09

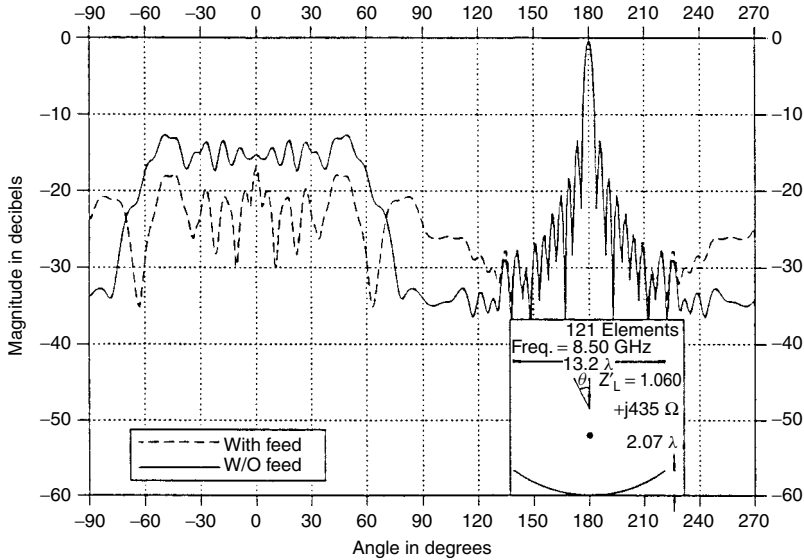


Fig. 8.10 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, $\theta = 0^\circ$. Shown with and without a low RCS feed. Frequency is 8.5 GHz.

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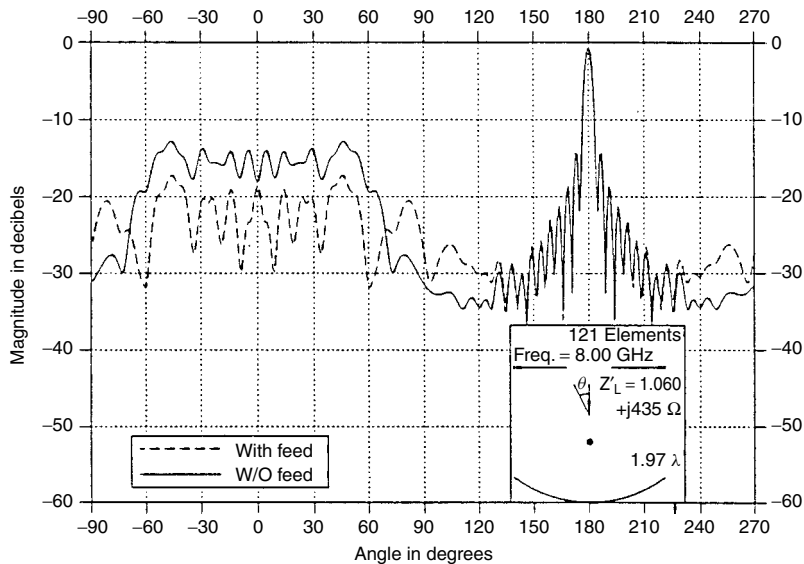


Fig. 8.11 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, $\theta = 0^\circ$. Shown with and without a low RCS feed. Frequency is 8.0 GHz.

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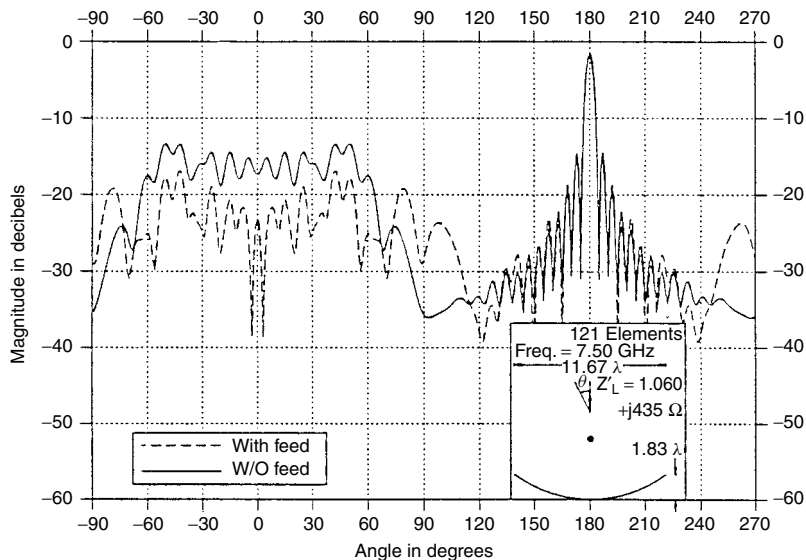


Fig. 8.12 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, $\theta = 0^\circ$. Shown with and without a low RCS feed. Frequency is 7.5 GHz.

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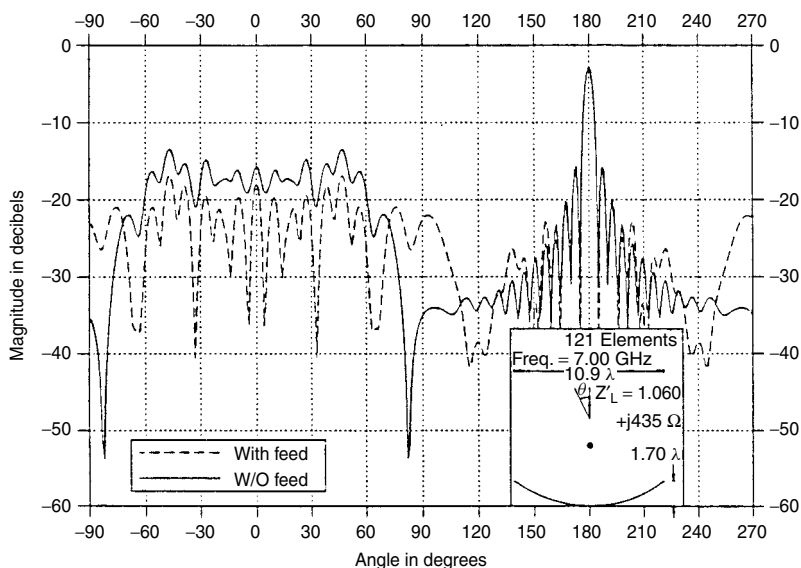


Fig. 8.13 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, $\theta = 0^\circ$. Shown with and without a low RCS feed. Frequency is 7.0 GHz.

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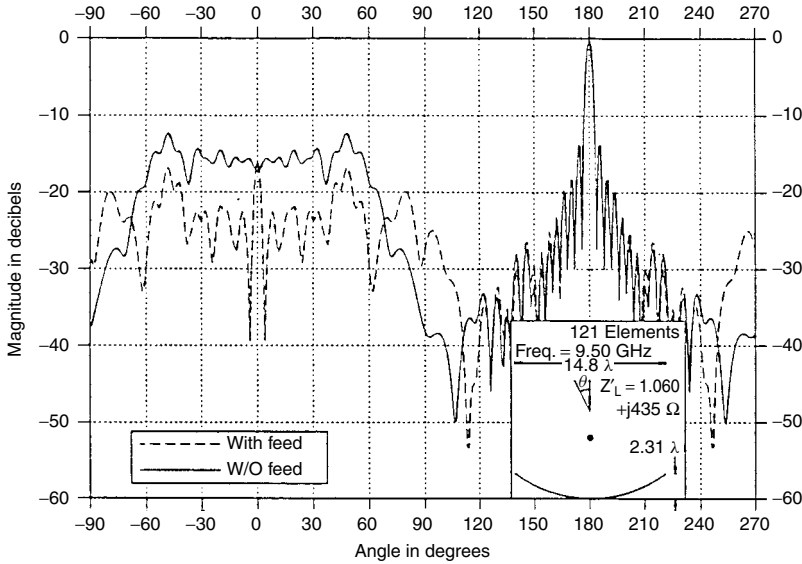


Fig. 8.14 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, $\theta = 0^\circ$. Shown with and without a low RCS feed. Frequency is 9.5 GHz.

Input file [.BDISH]BDISH10.PLT
 Number of arrays = 96
 Freq. = 1.0000000E + 09

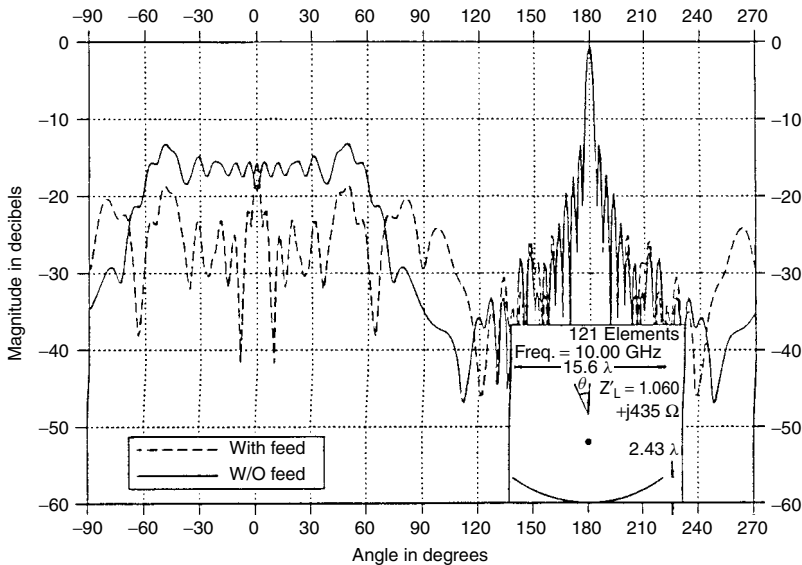


Fig. 8.15 The bistatic scattered field from a parabolic cylinder for an incident plane wave along boresight; that is, $\theta = 0^\circ$. Shown with and without a low RCS feed. Frequency is 10.0 GHz.

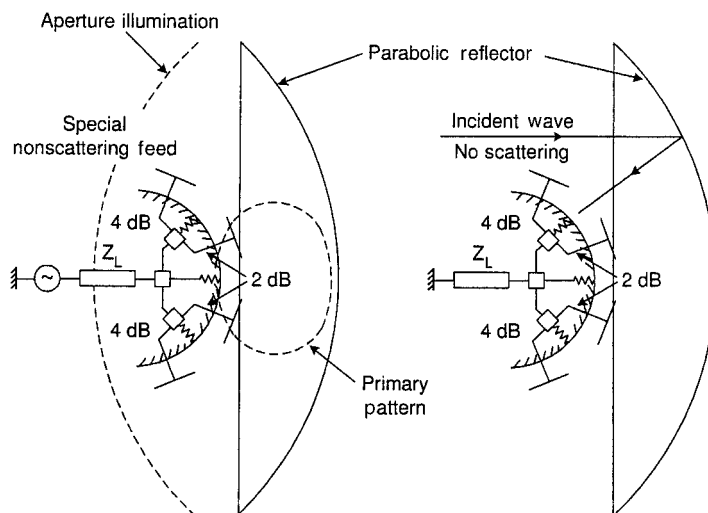


Fig. 8.16 A practical design of a low RCS feed comprised of dipoles in front of a circular groundplane and being fed via hybrids. Left: Transmitting case. The special $2/4$ dB hybrids will provide a tapered aperture distribution as desired. Right: Receiving case. All dipoles will be exposed to approximately the same power level. However, the hybrid will deliver the energy to the load impedance Z_L in a tapered fashion while excess power is being dissipated in the hybrid loads.

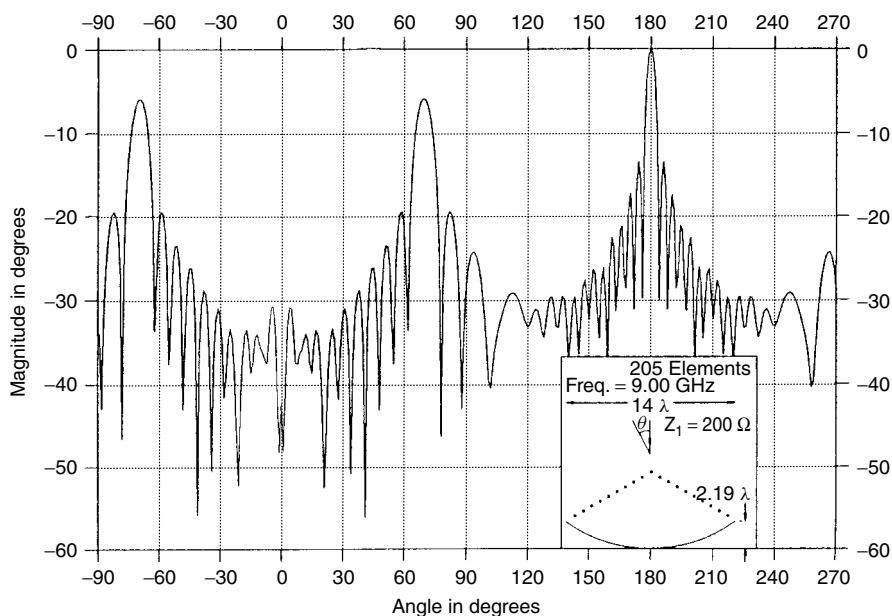


Fig. 8.17 Out-of-band reduction. An FSS radome covering the entire reflector aperture. The bistatic scattered field is obtained when the radome is opaque.

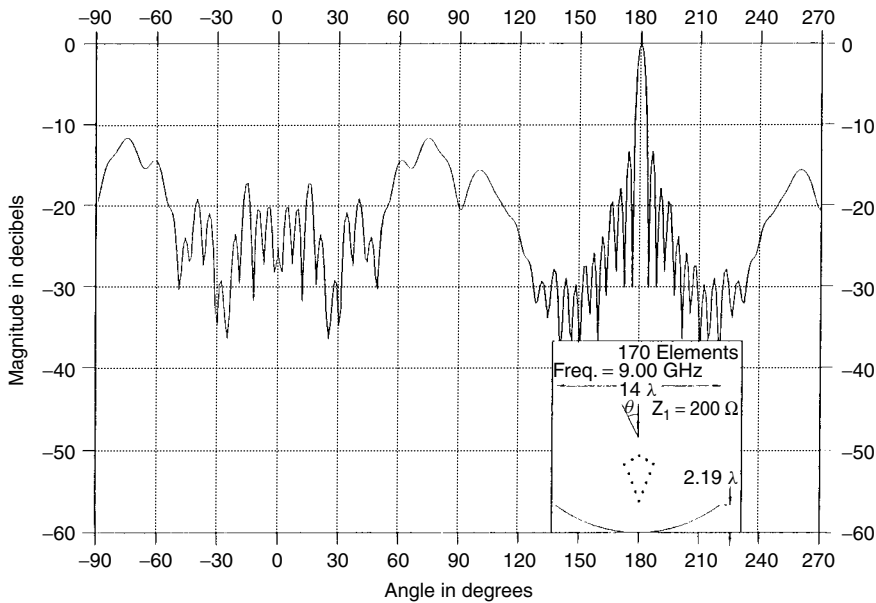


Fig. 8.18 Out-of-band reduction. An FSS radome surrounding the feed only. The bistatic scattered field is obtained when the radome is opaque. The wedge (or cone) angle is quite critical and so is the length of the radome. See text.

cone must be sufficiently long such that a convergent bundle of rays is reflected from it and not just a “big fat dipole pattern.” The reader is also reminded that the total transmitting energy will pass through the radome. Thus, consideration must be given to the power handling capacity in particular if the small feed radome is chosen. See also reference 110.

8.5 COMMON MISCONCEPTIONS ON EDGE CURRENTS

When discussing scattering from a parabolic antenna the question is often asked, “Did you incorporate the edge currents?” Our answer is, “Certainly!”

And the next question is, “Where was that done?”

Actually we calculated the bistatic scattering by direct application of the method of moments. It is therefore very accurate and includes everything. Other researchers may take a somewhat different approach. Basically they study the exact solution (like ours) and observe that it “looks like” one component is coming, for example, from the vertex of the dish, another from the edges, and still another from the feed itself, and so forth. These observations might indeed be correct and can even lead to a computationally very fast approach. However, one may also arrive at some erroneous conclusions. For example, it is often suggested that since a significant part of the reflector scattering seems to come from the

edges of the reflector, some form of edge treatment could reduce the reflector scattering. Although some minor change may be observed by attacking the edges only, it must be emphasized that any radiated field from any structure is derived from electrical (and/or magnetic) currents only and integrated over the *entire* body. As an illustration, consider our reflector model made of infinitely long FSS sticks located on a parabola (see Section 8.1.1). Any modification of the outermost FSS sticks would hardly change the currents in the rest of the reflecting elements and thus not the reflected field. In fact, if we ultimately simply removed the outermost sticks we would simply end up with a reflector slightly smaller than the original and we would be right back to square one. A tapered approach might be slightly better.

However, edge treatment is an entirely different matter if we instead of a single FSS surface had considered an array of conjugate-matched dipoles located in front of a groundplane as was the case in Chapter 5, Sections 5.3 and 5.4. If the edge columns were loaded with the same load as the rest of the array, namely the conjugate matched, we would indeed observe no reflection from the surface itself and observe a relatively strong scattering from the edge columns (see Figs. 5.2 and 5.3). This is simply because these columns are in an “outside” environment and thus have an antenna impedance quite different from the rest of the columns in the array. We can in that case load the outer columns with a load impedance being basically conjugate-matched to the outer columns. Note that these facts should not be interpreted as edges are always scattering. In fact, we can engineer surfaces that do not scatter at the edges but everywhere else. See Chapter 5 for details, as well as Problem 5.1.

8.6 CONCLUSION

We have demonstrated that the scattering from a parabolic cylindrical reflector consists essentially of two components: One is reflected from the reflector directly without the feed, and another is reflected from the feed via the reflector.

The first of these components, namely the reflector scattering, in general cannot be reduced significantly in band. Out of band a suitably designed radome placed either over the entire reflector or over the feed itself was as shown in Section 8.4 an effective way to reduce the frontal RCS.

The other component, namely the feed scattering, could be greatly reduced by using a new specially designed feed. It consisted, in principle, of dipole elements placed around a circular pipe acting as a “groundplane.” In fact, the bistatic return of the entire antenna was about 6 dB lower with the new feed than was the reflector scattering without any feed at all. The reason for this apparent paradox was that the field being reflected from the reflector would pass unperturbed through the focal area and thus be added to the field coming directly from the reflector. If these two components were approximately in phase, an increase in the total backscattered field is observed. If we instead use the low RCS feed, the field in the focal area would simply be absorbed and not continue through to the backscatter sector.

The RCS for a parabolic reflector cannot be as low as the planar array backed by a groundplane, see Chapters 2 and 5. However, they are discussed here because of their popularity in some camps.

Cassegrain antennas have also been investigated but are not reported here. The reason being that they exhibit certain problems relating to standing waves between the sub and main reflector. That makes the control of the backscatter difficult and they are therefore in general not recommended.

Our investigation here has primarily been focused around the two-dimensional cylindrical parabola. This case could be solved exactly and efficiently by using the SPLAT program. However, it is readily extended to explain the scattering mechanism for the parabolic disk as well.

Finally, I would like to point out that one of my former students, John Mehr, was extremely active in this investigation. Thank you John.